6 Qualified Manufacturing-Service Market

This section closes only the market for qualified manufacturing-service capacity. Developers and qualified suppliers take a conjectured price p_m as given when making their individual choices. The market-consistent price p_m^* is then selected by one scalar clearing equation. No entry, labor, capital, product-market, or welfare market is added.

6.1 Qualified-capacity supply

Qualified supplier j, with predetermined efficiency z_j, chooses capacity $s_j \geq 0$. Let $\Psi(s_j; z_j)$ denote its real capacity cost. The supplier solves

$$\max_{s_j \geq 0} \left\{ p_m s_j - \Psi(s_j; z_j) \right\}. \tag{1}$$

Capacity has units B per decision cohort, p_m has units C/B, and Ψ has units C.

For every z_j, assume twice continuous differentiability and

$$\Psi(0; z_j) = 0, \qquad \Psi_s(0; z_j) = 0,$$

$$\Psi_s > 0 \text{ for } s_j > 0, \qquad \Psi_{ss} > 0, \qquad \Psi_{sz} < 0 \text{ for } s_j > 0,$$

At zero capacity, $\Psi_{sz}(0; z_j) = 0$; no strict cross-derivative condition is imposed there. Assume $\Psi_s(s_j; z_j) \to +\infty$ as $s_j \to +\infty$. At $p_m = 0$, the unique optimum is $s_j^* = 0$. For $p_m > 0$, the solution is interior and satisfies

$$p_m = \Psi_s\Big(s_j^*(p_m, z_j); z_j\Big). \tag{2}$$

The objective's second derivative is $-\Psi_{ss} < 0$, so the capacity choice is unique. The implicit-function theorem gives

$$\frac{\partial s_j^*}{\partial p_m} = \frac{1}{\Psi_{ss}} > 0, \qquad \frac{\partial s_j^*}{\partial z_j} = -\frac{\Psi_{sz}}{\Psi_{ss}} > 0. \tag{3}$$

Aggregate qualified-capacity supply is

$$S_m(p_m) = \int s_j^*(p_m, z)\, dH_C(z). \tag{4}$$

Assume supplier best responses admit an integrable envelope on every bounded price interval and H_C has positive unit mass. Then S_m is continuous and strictly increasing, $S_m(0) = 0$, and $S_m(p_m) \to +\infty$ as $p_m \to +\infty$. Supply technology and H_C are invariant to M; MAH creates no direct supply shift.

6.2 Entrusted-capacity demand per planning-stage project

Assume $\int b(m)\, dF < \infty$, along with the Phase 5 value envelope and the Phase 4 no-tie restriction at each candidate price. At a candidate price p_m, define developer i's expected qualified capacity per planning-stage project as

$$\chi_i^E(p_m; M) = \int b(m) \mathbf{1}\left\{ r_i^*(q, m; M, p_m) = E \right\} dF(q, m). \tag{5}$$

This is an integral of deterministic route choices, not a probabilistic route-share formula.

The entrusted value falls with price at slope $-b(m) < 0$, while the other route values do not contain p_m. Consequently, for each project the set of prices at which E is chosen is an interval from below: once a higher price makes E lose to another route, a still higher price cannot restore it. Hence $\chi_i^E(p_m; M)$ is weakly decreasing in p_m.

Away from the measure-zero set of route ties, the envelope derivative of the optimized project value is

$$\frac{\partial W_i(q, m; M, p_m)}{\partial p_m} = -b(m)\mathbf{1}\{r_i^*(q, m; M, p_m) = E\}.$$

Dominated convergence then gives

$$\frac{\partial \Omega_i(M, p_m)}{\partial p_m} = -\chi_i^E(p_m; M) \leq 0 \quad \text{almost everywhere in } p_m. \tag{6}$$

For $\Omega_i > 0$, the advancement optimizer therefore satisfies

$$\frac{\partial x_i^*(M, p_m)}{\partial p_m} = -\frac{x_i^*(M, p_m)}{\nu \Omega_i(M, p_m)} \chi_i^E(p_m; M) \leq 0. \tag{7}$$

At the zero-value corner, $x_i^* = 0$; globally the optimal advancement intensity remains weakly decreasing in p_m.

6.3 Aggregate demand

Developer i's expected qualified-capacity demand is

$$a_i x_i^*(M, p_m) \chi_i^E(p_m; M).$$

Aggregating over developer heterogeneity gives study-related demand

$$D_m^{\mathrm{MAH}}(p_m; M) = \int a_i x_i^*(M, p_m) \chi_i^E(p_m; M) dH(a, k). \tag{8}$$

This expression contains both price feedbacks required by the model: x_i^* responds through expected project value, and χ_i^E responds through deterministic route selection.

For aggregate integration, assume $\int a_i x_i^*(1, 0)[\int b(m)\, dF]\, dH < \infty$. This is a sufficient common envelope for all nonnegative candidate prices and both regimes because advancement and entrusted demand are nonincreasing in price and pre-reform advancement is no greater than post-reform advancement at zero price. Both factors are nonnegative and weakly decreasing in p_m, so D_m^{MAH} is weakly decreasing. Individual route indicators can jump at a cutoff. Aggregate demand is nevertheless continuous: when $p_m^n \to p_m$, route indicators converge for every project–developer pair except the zero-measure tie set, x_i^* is continuous through Ω_i, and the common integrable envelope permits dominated convergence. Thus aggregate regularity follows from the no-tie and integrability restrictions on heterogeneity; continuity of heterogeneity alone is insufficient.

Let $D_m^B(p_m)$ be finite, continuous, nonnegative, weakly decreasing background demand with

$$D_m^B(0) > 0, \qquad \lim_{p_m \to \infty} D_m^B(p_m) = 0.$$

It is policy invariant and allows a CMO market to exist before MAH. Total demand is

$$D_m(p_m; M) = D_m^B(p_m) + D_m^{\mathrm{MAH}}(p_m; M). \tag{9}$$

It is finite, continuous, nonnegative, and weakly decreasing.

6.4 Market clearing, existence, and uniqueness

The qualified-capacity market clears at

$$D_m(p_m^*; M) = S_m(p_m^*). \tag{10}$$

At zero price,

$$D_m(0; M) \geq D_m^B(0) > 0 = S_m(0). \tag{11}$$

As $p_m \to \infty$, $W_i^E \to -\infty$ pointwise because $b(m) > 0$, so $\chi_i^E \to 0$ and study-related demand tends to zero under the integrable envelope. Background demand also tends to zero, whereas aggregate supply tends to infinity. Continuity therefore gives at least one positive clearing price.

Total demand is weakly decreasing and aggregate supply is strictly increasing. Their difference is strictly decreasing, so the clearing price is unique.

6.5 Closed scalar solution order

For each regime M, the equilibrium can be computed without an unresolved cycle:

1. conjecture p_m;
2. evaluate route values and deterministic $r_i^*(q, m; M, p_m)$;
3. integrate $\Omega_i(M, p_m)$ and compute $x_i^*(M, p_m)$;
4. integrate χ_i^E and D_m^{MAH};
5. solve supplier capacity and aggregate $S_m(p_m)$;
6. find the unique zero of $D_m(p_m; M) - S_m(p_m)$.

Every object on the right side is a function of the candidate price, leaving one scalar equation in p_m.

When $M = 0$, route E is unavailable, so $D_m^{\mathrm{MAH}}(p_m; 0) = 0$; background demand still supports the market. When $M = 1$, study demand is nonnegative and supply is unchanged. Hence the post-MAH clearing price is weakly above the pre-MAH price, and it is strictly higher if study demand is positive at the pre-MAH price. This is an endogenous scarcity-price response, not a direct policy shift in p_m^* or CMO technology.

6.6 Scope boundary

This section closes qualified manufacturing-service capacity only. It introduces no supplier entry, no direct policy supply shift, no logit route smoothing, and no additional market-clearing condition. The price response is generated jointly by supplier optimization, deterministic route selection, project advancement, and scalar market clearing.